The Coh Category V2

Projection-Aware Certification and Oplax Semantic Cost

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Abstract. We define the second version of the Coh category, a projection-aware categorical structure in which morphisms are generated by certified observable traces rather than assumed as primitive arrows. The primitive layer consists of hidden realizations, observable traces, a projection map, and a hidden cost functional. Observable cost is not read directly from a trace; it is defined as an envelope defect, the supremum of hidden cost over the fiber of all hidden realizations projecting to the same observable record. We prove envelope soundness, envelope minimality, identity cost, projection-induced subadditivity, closure of certified composition, identity certification, associativity, local hom-preorder structure, and a directed Lawvere-style distance. The central structural law is the oplax inequality

$$\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2),$$

which expresses the fact that observable composition can only be certified through worst-case hidden fibers. The result is a verifier-ready mathematical foundation for resource-bounded execution, projection-safe semantics, and later Lean formalization.

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1 Introduction

In ordinary categorical modeling, arrows are usually part of the input data. A set of objects is specified, a class of morphisms is supplied, and additional structure such as cost, weight, probability, or enrichment is attached afterward. That order is not appropriate for verification-sensitive computation. In a governed system, a transition should not count as an available morphism merely because a source and target can be named. It must first be certified.

The Coh construction reverses the order:

$$\text{proposal} \quad \longrightarrow \quad \text{verification} \quad \longrightarrow \quad \text{morphism}.$$

Version 1 of the Coh category isolates the operational rule: a trace is admissible when it satisfies verifier acceptance and the resource inequality

$$V(y) + \mathrm{Spend}(\tau) \leq V(x) + \mathrm{Defect}(\tau). \tag{1}$$

Version 2 adds the missing projection layer. Observable traces may hide many possible underlying realizations. Therefore an observable trace cannot be assigned semantic cost by inspecting the observable record alone. It must be assigned a conservative envelope over its hidden fiber.

The V2 foundation is built around the following spine:

hidden realizations → observable projection → fiber envelope → certified morphisms → category.

The central object is the envelope defect

$$\delta(R) := \sup_{\xi \in \mathcal{F}_R} W(\xi), \qquad \mathcal{F}_R := \{\xi \in \mathcal{G} : \Pi(\xi) = R\}. \tag{2}$$

This enforces the V2 rule: every observable claim must be valid for the worst hidden realization consistent with the observable receipt.

Contributions

The paper proves the following results from explicit axioms.

(1) **Envelope soundness.** Every hidden realization is bounded by the envelope of its observable projection.

(2) **Envelope minimality.** The envelope is the least conservative bound that dominates hidden cost fiberwise.

(3) **Oplax subadditivity.** Projection induces

$$\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2).$$

(4) **Certified category theorem.** Endpoint-typed certified morphisms compose and define a category.

(5) **Local preorder and metric enrichment.** Hom-sets carry a preorder by declared defect, and states carry a directed extended distance obtained by infimizing envelope defect.

Status of the claims

All theorem statements below are conditional on the listed axioms. No physical universality claim is made. The paper proves a structural theorem: if a system supplies the specified hidden layer, projection, decomposition, and cost rules, then the certified observable layer has the stated categorical and oplax properties.

Table 1. Dictionary from V1 operational terminology to V2 projection-aware terminology.

Layer	V1 term	V2 term
Observable record	trace τ	observable trace $R \in \mathcal{V}$
Hidden semantics	implicit execution	hidden realization $\xi \in \mathcal{G}$
Projection	not first-class	$\Pi : \mathcal{G} \to \mathcal{V}$
Semantic cost	optional realized cost	envelope $\delta(R) = \sup_{\xi \in \mathcal{F}_R} W(\xi)$
Certification	$V(y) + S \leq V(x) + D$	$d \geq \delta(R)$ and $V(y) + s \leq V(x) + d$
Composition	concatenation	partial composition $R_2 \odot R_1$
Structural law	additive bookkeeping	oplax semantic bound

2 V1-to-V2 bridge

V1 is the operational category of admissible traces. V2 is the projection-aware version of that category. The bridge is summarized in table 1.

The important change is that V2 does not reason directly on observable traces. It reasons on the fiber of hidden realizations compatible with the observable trace. Thus the semantic meaning of an observable record is not a selected hidden realization; it is the worst-case envelope over all hidden realizations projecting to that record.

3 Primitive projection system

3.1 Objects and operators

Definition 3.1 (Hidden substrate)**.** A hidden trajectory substrate consists of a class or set $\mathcal{G}$ of hidden realizations and a partially defined concatenation operation

$$\cdot : \mathcal{G} \times \mathcal{G} \rightharpoonup \mathcal{G}.$$

When $\xi_2 \cdot \xi_1$ is defined, ξ_1 is executed first and ξ_2 second.

Definition 3.2 (Observable substrate)**.** An observable trace substrate consists of a set $\mathcal{V}$ of observable traces, a partially defined concatenation operation

$$\odot : \mathcal{V} \times \mathcal{V} \rightharpoonup \mathcal{V},$$

and a distinguished identity trace $e \in \mathcal{V}$.

Definition 3.3 (Projection and hidden cost)**.** A projection-cost layer consists of a map

$$\Pi : \mathcal{G} \to \mathcal{V}$$

and a nonnegative hidden cost functional

$$W : \mathcal{G} \to \mathbb{R}_{\geq 0}.$$

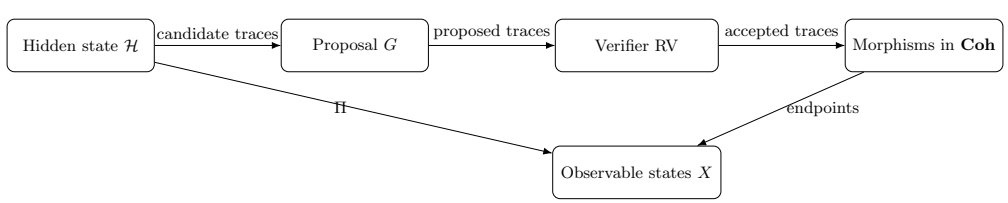


Figure 1. Execution wedge. Hidden-state proposals generate candidate traces, the verifier certifies a subset, and only certified traces enter the categorical layer. Projection connects hidden realizations to observable receipts.

3.2 Structural axioms

Axiom 3.4 (Observable associativity)**.** For all composable $R_1, R_2, R_3 \in \mathcal{V}$,

$$(R_3 \odot R_2) \odot R_1 = R_3 \odot (R_2 \odot R_1).$$

Axiom 3.5 (Observable identity)**.** For every $R \in \mathcal{V}$,

$$e \odot R = R, \qquad R \odot e = R,$$

whenever either side is defined.

Axiom 3.6 (Projection compatibility)**.** If $\xi_2 \cdot \xi_1$ is defined, then

$$\Pi(\xi_2 \cdot \xi_1) = \Pi(\xi_2) \odot \Pi(\xi_1).$$

Axiom 3.7 (Fiber decomposition)**.** If $\xi \in \mathcal{G}$ satisfies

$$\Pi(\xi) = R_2 \odot R_1,$$

then there exist $\xi_1, \xi_2 \in \mathcal{G}$ such that

$$\xi = \xi_2 \cdot \xi_1, \qquad \Pi(\xi_1) = R_1, \qquad \Pi(\xi_2) = R_2.$$

Axiom 3.8 (Additive hidden cost)**.** If $\xi_2 \cdot \xi_1$ is defined, then

$$W(\xi_2 \cdot \xi_1) = W(\xi_1) + W(\xi_2).$$

Axiom 3.9 (Identity fiber)**.** The identity fiber is nonempty and zero-cost:

$$\Pi^{-1}(e) \neq \varnothing, \qquad \forall \xi \in \Pi^{-1}(e),\ W(\xi) = 0.$$

Remark 3.10 (Computability boundary)*.* The abstract functional δ below is stated using a supremum. In an implementation, this supremum is replaced by a deterministic conservative upper envelope. The proof obligation is not that the verifier computes the exact supremum; it is that the declared defect dominates a valid upper bound for the hidden fiber.

4 Envelope defect

Definition 4.1 (Hidden fiber)**.** For $R \in \mathcal{V}$, define the hidden fiber of R by

$$\mathcal{F}_R := \{\xi \in \mathcal{G} : \Pi(\xi) = R\}.$$

Definition 4.2 (Admissible observable trace)**.** An observable trace $R \in \mathcal{V}$ is projection-admissible if $\mathcal{F}_R \neq \varnothing$. Write

$$\mathcal{V}_{\mathrm{adm}} := \{R \in \mathcal{V} : \mathcal{F}_R \neq \varnothing\}.$$

Axiom 4.3 (Fiber-boundedness)**.** For every $R \in \mathcal{V}_{\mathrm{adm}}$,

$$\sup_{\xi \in \mathcal{F}_R} W(\xi) < \infty.$$

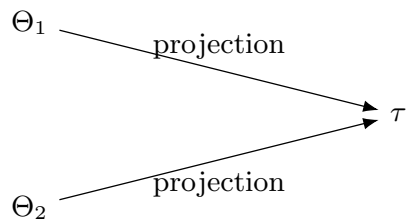
Definition 4.4 (Envelope defect)**.** For $R \in \mathcal{V}_{\mathrm{adm}}$, define the projection-induced envelope defect

$$\delta(R) := \sup_{\xi \in \mathcal{F}_R} W(\xi).$$

Lemma 4.5 (Envelope soundness)**.** *For every $\xi \in \mathcal{G}$ with $\Pi(\xi) \in \mathcal{V}_{\mathrm{adm}}$,*

$$W(\xi) \leq \delta(\Pi(\xi)).$$

Proof. Let $R = \Pi(\xi)$. By definition of fiber, $\xi \in \mathcal{F}_R$. Since $\delta(R)$ is the supremum of W over $\mathcal{F}_R$, it is an upper bound for all values $W(\eta)$ with $\eta \in \mathcal{F}_R$. In particular, $W(\xi) \leq \delta(R) = \delta(\Pi(\xi))$. $\square$



multiple hidden realizations can collapse to one observable trace

Figure 2. Projection collapses hidden realizations to observable traces. V2 certification must therefore use the worst-case hidden cost over the fiber, not a selected representative.

Lemma 4.6 (Envelope minimality)**.** *Let* $\Delta : \mathcal{V}_{\mathrm{adm}} \to \mathbb{R}_{\geq 0}$ *satisfy*

$$W(\xi) \leq \Delta(\Pi(\xi))$$

for every hidden realization ξ *whose projection is admissible. Then for every* $R \in \mathcal{V}_{\mathrm{adm}}$*,*

$$\delta(R) \leq \Delta(R).$$

Proof. Fix $R \in \mathcal{V}_{\mathrm{adm}}$. For each $\xi \in \mathcal{F}_R$, one has $\Pi(\xi) = R$, hence

$$W(\xi) \leq \Delta(\Pi(\xi)) = \Delta(R).$$

Thus $\Delta(R)$ is an upper bound for the set $\{W(\xi) : \xi \in \mathcal{F}_R\}$. Since $\delta(R)$ is the least upper bound of that set, $\delta(R) \leq \Delta(R)$. □

Lemma 4.7 (Identity envelope)**.** *The identity trace satisfies*

$$\delta(e) = 0.$$

Proof. By axiom 3.9, $\mathcal{F}_e = \Pi^{-1}(e)$ is nonempty and every element in this fiber has cost zero. Therefore the set of costs over the identity fiber is $\{0\}$. Its supremum is zero. □

5 Projection-induced oplax composition

The next theorem is the central V2 result. It is not an optional cost estimate; it is the structural reason that observable composition is sound under hidden projection.

Theorem 5.1 (Projection-induced subadditivity)**.** *Let* $R_1, R_2 \in \mathcal{V}_{\mathrm{adm}}$*, and assume* $R_2 \odot R_1 \in \mathcal{V}_{\mathrm{adm}}$*. Then*

$$\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2). \tag{3}$$

Proof. Let $\xi \in \mathcal{F}_{R_2 \odot R_1}$. By definition 4.1,

$$\Pi(\xi) = R_2 \odot R_1.$$

By fiber decomposition, there exist $\xi_1, \xi_2 \in \mathcal{G}$ such that

$$\xi = \xi_2 \cdot \xi_1, \qquad \Pi(\xi_1) = R_1, \qquad \Pi(\xi_2) = R_2.$$

Thus $\xi_i \in \mathcal{F}_{R_i}$ for $i = 1, 2$. By additive hidden cost,

$$W(\xi) = W(\xi_2 \cdot \xi_1) = W(\xi_1) + W(\xi_2).$$

By envelope soundness,

$$W(\xi_1) \leq \delta(R_1), \qquad W(\xi_2) \leq \delta(R_2).$$

Therefore

$$W(\xi) \leq \delta(R_1) + \delta(R_2).$$

Since ξ was arbitrary in $\mathcal{F}_{R_2 \odot R_1}$, the right-hand side is an upper bound for all hidden costs over the composite fiber. Taking the supremum gives (3). □

Remark 5.2 (Oplax, not additive). The theorem gives an inequality, not an equality. Projection may merge hidden realizations, lose information, or permit hidden reuse that is not visible at the observable layer. The correct observable law is therefore subadditive.

6 Endpoint-typed certified morphisms

Definition 6.1 (Coh boundary)**.** A Coh boundary object X consists of a carrier set Σ_X and a valuation

$$V_X : \Sigma_X \to \mathbb{R}_{\geq 0}.$$

Definition 6.2 (Endpoint typing)**.** For boundary objects X, Y, an admissible observable trace $R \in \mathcal{V}_{\mathrm{adm}}$ is compatible with endpoints $x \in \Sigma_X$ and $y \in \Sigma_Y$ if it carries a typing relation

$$R : x \rightsquigarrow y.$$

This relation is primitive at the V2 foundation layer.

Definition 6.3 (Certified morphism)**.** Let X, Y be Coh boundary objects. A certified morphism

$$f : X \to Y$$

is a tuple

$$f = (R_f, s_f, d_f, x, y)$$

such that:

(1) $x \in \Sigma_X$ and $y \in \Sigma_Y$;
(2) $R_f \in \mathcal{V}_{\mathrm{adm}}$;
(3) $R_f : x \rightsquigarrow y$;
(4) the declared defect dominates the projection envelope,

$$d_f \geq \delta(R_f);$$

(5) the endpoint resource inequality holds,

$$V_Y(y) + s_f \leq V_X(x) + d_f.$$

Here $s_f \in \mathbb{R}_{\geq 0}$ is declared spend and $d_f \in \mathbb{R}_{\geq 0}$ is declared defect.

Remark 6.4 (Separation of spend and defect). Spend and defect have different roles. Spend is paid transition cost. Defect is declared tolerance, uncertainty, or projection allowance. In V2, defect also must dominate the semantic envelope δ.

Definition 6.5 (Certified composition)**.** Let

$$f = (R_f, s_f, d_f, x, y) : X \to Y, \qquad g = (R_g, s_g, d_g, y, z) : Y \to Z.$$

Assume $R_g \odot R_f$ is defined, admissible, and typed

$$R_g \odot R_f : x \rightsquigarrow z.$$

Define

$$g \circ f := (R_g \odot R_f,\ s_f + s_g,\ d_f + d_g,\ x,\ z).$$

Definition 6.6 (Identity morphism)**.** For each Coh boundary object X and each $x \in \Sigma_X$, define

$$\mathrm{id}_x := (e, 0, 0, x, x).$$

$$\begin{tikzcd} x \arrow[r, "\tau_1"] \arrow[rr, bend left, "\tau_2 \circ \tau_1 = \tau_2 * \tau_1"] & y \arrow[r, "\tau_2"] & z \end{tikzcd}$$

Figure 3. Certified composition. If $R_1 : x \rightsquigarrow y$ and $R_2 : y \rightsquigarrow z$ are certified and compatible, the composite observable trace is $R_2 \odot R_1 : x \rightsquigarrow z$.

7 Certified category theorem

Lemma 7.1 (Closure under certified composition)**.** *Let $f : X \to Y$ and $g : Y \to Z$ be certified morphisms as in definition 6.5. If the composite observable trace is defined, admissible, and correctly typed, then $g \circ f$ is certified.*

Proof. The endpoint and typing requirements hold by the hypothesis of certified composition. It remains to prove the envelope dominance and resource inequality.

By theorem 5.1,

$$\delta(R_g \odot R_f) \leq \delta(R_f) + \delta(R_g).$$

Since f and g are certified,

$$d_f \geq \delta(R_f), \qquad d_g \geq \delta(R_g).$$

Adding yields

$$d_f + d_g \geq \delta(R_f) + \delta(R_g) \geq \delta(R_g \odot R_f).$$

Thus the declared defect of the composite dominates the envelope of the composite trace.

For the endpoint resource inequality, certification gives

$$V_Y(y) + s_f \leq V_X(x) + d_f,$$

and

$$V_Z(z) + s_g \leq V_Y(y) + d_g.$$

Adding the inequalities gives

$$V_Y(y) + V_Z(z) + s_f + s_g \leq V_X(x) + V_Y(y) + d_f + d_g.$$

Cancel the finite nonnegative term $V_Y(y)$ to obtain

$$V_Z(z) + (s_f + s_g) \leq V_X(x) + (d_f + d_g).$$

This is exactly the certified resource inequality for $g \circ f$. □

Lemma 7.2 (Certified identities)**.** *For every boundary object X and every $x \in \Sigma_X$, the tuple $\mathrm{id}_x = (e, 0, 0, x, x)$ is certified.*

Proof. By axiom 3.9, the identity trace is admissible. By lemma 4.7, $\delta(e) = 0$, so the defect condition $0 \geq \delta(e)$ holds. The resource inequality is

$$V_X(x) + 0 \leq V_X(x) + 0,$$

which holds by equality. Thus id_x is certified. □

Lemma 7.3 (Associativity)**.** *Let f, g, h be composable certified morphisms, and assume all relevant composite traces are defined, admissible, and correctly typed. Then*

$$(h \circ g) \circ f = h \circ (g \circ f).$$

Proof. By definition of certified composition, the trace component of $(h \circ g) \circ f$ is $(R_h \odot R_g) \odot R_f$, while the trace component of $h \circ (g \circ f)$ is $R_h \odot (R_g \odot R_f)$. These are equal by observable associativity. The spend components are equal by associativity of addition:

$$(s_f + s_g) + s_h = s_f + (s_g + s_h).$$

The defect components are equal for the same reason. Source and target endpoints agree by composability. Hence the certified tuples are equal. □

Theorem 7.4 (The Coh Category V2)**.** *Certified boundary states and certified morphisms form a category, denoted* Coh_δ*.*

Proof. Closure under composition is lemma 7.1. Certified identities are provided by lemma 7.2. Associativity is lemma 7.3. The unit laws follow from the observable identity law and the fact that adding zero spend and zero defect leaves the certified tuple unchanged. Therefore the certified system forms a category. □

8 Local preorder by defect

Definition 8.1 (Hom-set preorder)**.** For certified morphisms $f, g \in \mathsf{Coh}_\delta(X, Y)$, define

$$f \preceq g \quad \Longleftrightarrow \quad d_f \leq d_g.$$

Proposition 8.2 (Preorder)**.** *For each fixed pair of boundary objects X, Y, the relation $\preceq$ is a preorder on* $\mathsf{Coh}_\delta(X, Y)$*.*

Proof. Reflexivity follows from $d_f \leq d_f$. Transitivity follows because $d_f \leq d_g$ and $d_g \leq d_h$ imply $d_f \leq d_h$. □

Proposition 8.3 (Monotonicity of composition)**.** *If $f_1 \preceq f_2$ and $g_1 \preceq g_2$, and both composites are defined, then*

$$g_1 \circ f_1 \preceq g_2 \circ f_2.$$

Proof. The assumptions mean $d_{f_1} \leq d_{f_2}$ and $d_{g_1} \leq d_{g_2}$. Adding gives

$$d_{f_1} + d_{g_1} \leq d_{f_2} + d_{g_2}.$$

The defect component of a composite is the sum of the defect components. Therefore the defect of $g_1 \circ f_1$ is no larger than the defect of $g_2 \circ f_2$, which is precisely the stated preorder relation. □

9 Metric enrichment

The envelope defect induces a directed extended distance by minimizing certified semantic cost over admissible traces.

Definition 9.1 (Coh distance)**.** For endpoint states x and y, define

$$d_{\mathsf{Coh}_\delta}(x, y) := \inf\{\delta(R) : R \in \mathcal{V}_{\mathrm{adm}},\ R : x \rightsquigarrow y\}.$$

If no admissible trace exists from x to y, set $d_{\mathsf{Coh}_\delta}(x, y) = +\infty$.

Lemma 9.2 (Zero self-distance)**.** *For every endpoint state x,*

$$d_{\mathsf{Coh}_\delta}(x, x) = 0.$$

Proof. The identity trace $e : x \rightsquigarrow x$ is admissible, and $\delta(e) = 0$. Therefore $d_{\mathsf{Coh}_\delta}(x, x) \leq 0$. Since all envelope defects are nonnegative, $d_{\mathsf{Coh}_\delta}(x, x) \geq 0$. Hence equality holds. □

Theorem 9.3 (Directed triangle inequality)**.** *For endpoint states x, y, z,*

$$d_{\mathsf{Coh}_\delta}(x, z) \leq d_{\mathsf{Coh}_\delta}(x, y) + d_{\mathsf{Coh}_\delta}(y, z).$$

Proof. If either term on the right is $+\infty$, the inequality is immediate. Otherwise fix $\varepsilon > 0$. By definition of infimum, choose admissible traces $R_1 : x \rightsquigarrow y$ and $R_2 : y \rightsquigarrow z$ such that

$$\delta(R_1) \leq d_{\mathsf{Coh}_\delta}(x, y) + \varepsilon/2, \qquad \delta(R_2) \leq d_{\mathsf{Coh}_\delta}(y, z) + \varepsilon/2.$$

By certified closure, $R_2 \odot R_1 : x \rightsquigarrow z$ is admissible whenever the composition hypotheses are satisfied. By theorem 5.1,

$$\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2) \leq d_{\mathsf{Coh}_\delta}(x, y) + d_{\mathsf{Coh}_\delta}(y, z) + \varepsilon.$$

Since $d_{\mathsf{Coh}_\delta}(x, z)$ is the infimum over admissible traces from x to z,

$$d_{\mathsf{Coh}_\delta}(x, z) \leq d_{\mathsf{Coh}_\delta}(x, y) + d_{\mathsf{Coh}_\delta}(y, z) + \varepsilon.$$

Letting $\varepsilon \to 0$ proves the claim. □

Corollary 9.4 (Lawvere-style enrichment)**.** *The state-indexed Coh category carries a directed extended pseudometric. Equivalently, it admits a Lawvere-style metric enrichment over the monoidal poset* $([0, \infty], \geq, +, 0)$.

Proof. Nonnegativity follows from nonnegativity of W and hence of δ. Zero self-distance is lemma 9.2. The directed triangle inequality is theorem 9.3. Symmetry is not assumed and generally fails. These are exactly the data of a directed extended pseudometric, equivalently a Lawvere metric structure. □

10 Worked separation example

V2 separates syntactic bookkeeping from semantic envelope cost. The following example records the distinction without relying on any physical interpretation.

Example 10.1 (Strict semantic compression)**.** Let $X = \{A, B, C\}$ and let a two-step observable trace be

$$A \xrightarrow{\sigma_1} B \xrightarrow{\sigma_2} C.$$

Suppose syntactic step spends satisfy

$$s(\sigma_1) = 1, \qquad s(\sigma_2) = 1,$$

so the syntactic spend of the composite is 2. Suppose the hidden fiber of the composite observable trace contains a single verifier-realizable hidden representative Θ_* with

$$W(\Theta_*) = 1.$$

Then

$$\delta(\sigma_2 \odot \sigma_1) = \sup_{\Theta \in \{\Theta_*\}} W(\Theta) = 1 < 2.$$

Thus the semantic envelope of the composite can be strictly smaller than stepwise syntactic bookkeeping.

Remark 10.2 (Meaning of the example)*.* The example does not say that bookkeeping is wrong. It says that V2 separates a safe syntactic ledger from a projection-induced semantic envelope. The declared defect channel must dominate the semantic envelope even when the syntactic ledger records a different cost channel.

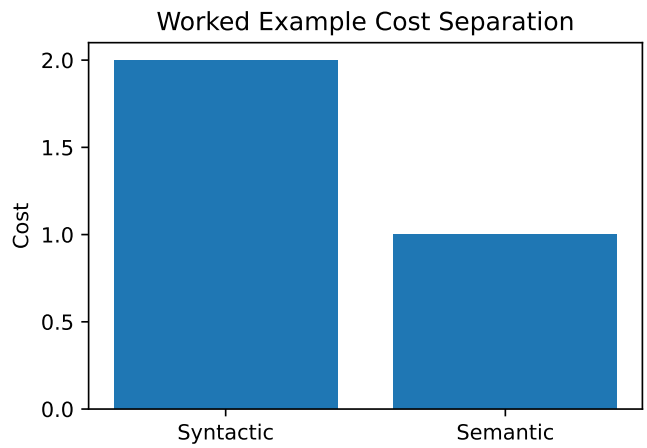


Figure 4. Worked cost separation. Syntactic bookkeeping can overcount realized semantic cost after projection through hidden realizations.

11 Lean and verifier alignment

The formal spine is intentionally type-safe. A Lean implementation can mirror the V2 construction with the following dependency order:

`Primitive` → `Axioms` → `Envelope` → `Certified` → `Category` → `Metric`.

A certified morphism should be encoded as a dependent record containing:

(1) endpoint states and boundary objects;
(2) an admissible observable trace;
(3) endpoint typing evidence;
(4) a proof that declared defect dominates the envelope;
(5) a proof of the endpoint resource inequality.

At the computation boundary, real-valued suprema should not be trusted as executable objects. The verifier should accept a deterministic, versioned, finite representation of a conservative envelope. The theorem obligation is:

$$\widehat{\delta}(R) \geq \delta(R),$$

where $\widehat{\delta}$ is the computable certificate used in the receipt. Certification then requires

$$d_f \geq \widehat{\delta}(R_f),$$

which implies $d_f \geq \delta(R_f)$ by transitivity.

12 Failure modes and boundary cases

12.1 Unbounded fibers

If fiber-boundedness fails, $\delta(R)$ may be infinite. The categorical construction can be extended-valued, but finite certification requires a declared finite upper envelope.

12.2 Projection failure

If the observable record omits information needed to construct a conservative fiber envelope, then the verifier may certify a surrogate weaker than the hidden system. This is not a failure of the category theorem; it is a failure of the projection design.

12.3 Missing fiber decomposition

The proof of subadditivity depends on decomposing hidden realizations of a composite observable trace into hidden realizations of its components. If this axiom fails, the oplax theorem is not available.

12.4 Over-strict admissibility

If the verifier rejects nearly all non-identity traces, the category degenerates toward identity morphisms. The theory remains valid, but the reachable geometry becomes trivial.

12.5 Loose defect

If declared defect is allowed to dominate every hidden cost without discipline, the category becomes broad but loses governing force. The mathematical structure survives, but the system no longer enforces meaningful selectivity.

13 Conclusion

The Coh Category V2 is a category of earned morphisms under projection-aware certification. Hidden realizations project to observable traces, observable traces receive semantic cost through a worst-case fiber envelope, and certified morphisms must declare enough defect to dominate that envelope while satisfying the endpoint resource inequality.

The central theorem is the oplax law

$$\delta(R_2 \odot R_1) \leq \delta(R_1) + \delta(R_2),$$

which converts hidden additive cost into observable subadditive semantic cost. From this law, certified composition closes, identities certify, hom-sets inherit a preorder by defect, and endpoint states carry a directed extended distance.

The safe structural claim is therefore:

Projection-aware admissibility generates a certified category with oplax semantic cost.

This is the V2 foundation on which later governed dynamics, metric enrichment, and executable verifier packages can be built.

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